

Q1
THE CRATE HAS WEIGHT DOWN, THE NORMAL FORCE OUT OF THE RAMP, AND FRICTION UP THE SLOPE, BECAUSE NOTHING ELSE IS TOUCHING IT, AND THIS HOLDS AS LONG AS THE RAMP SURFACE STAYS IN CONTACT. THE NORMAL FORCE IS PERPENDICULAR TO THE RAMP AND FRICTION IS PARALLEL TO IT, BECAUSE CONTACT FORCES SPLIT INTO THOSE TWO DIRECTIONS, AND THIS HOLDS AS LONG AS THE CRATE IS NOT TIPPING. THE CRATE STAYS PUT BECAUSE FRICTION BALANCES THE WEIGHT COMPONENT ALONG THE RAMP, BECAUSE THE TWO ALONG-RAMP FORCES ARE EQUAL AND OPPOSITE, AND THIS HOLDS AS LONG AS THE REQUIRED FRICTION STAYS UNDER THE LIMITING VALUE.

Q2

I SWUNG EQUAL ARCS FROM A AND FROM B AND JOINED THE TWO CROSSINGS, BECAUSE THE CROSSINGS ARE EQUIDISTANT FROM BOTH ENDPOINTS, AND THIS HOLDS AS LONG AS THE RADIUS IS MORE THAN HALF OF AB. THE ARCS MEET AT P AND Q, AND PA EQUALS PB AND QA EQUALS QB, BECAUSE EACH ARC WAS DRAWN WITH ONE RADIUS, AND THIS HOLDS AS LONG AS THE COMPASS IS NOT RESET BETWEEN ARCS.

PQ IS PERPENDICULAR TO AB AND CUTS IT IN HALF, BECAUSE P AND Q ARE BOTH EQUIDISTANT FROM A AND B, AND THIS HOLDS AS LONG AS A AND B ARE DISTINCT POINTS.

Q3

THE HORIZONTAL AXIS IS TIME IN SECONDS AND THE VERTICAL AXIS IS VELOCITY IN METRES PER SECOND, BECAUSE THE AXIS LABELS CARRY THEIR UNITS, AND THIS HOLDS AS LONG AS THE SCALE IS LINEAR ON BOTH AXES.

IT SPEEDS UP FOR FOUR SECONDS, TURNS AROUND AT THE PEAK, THEN SLOWS TO REST, BECAUSE THE CURVE RISES, REACHES A MAXIMUM, THEN FALLS, AND THIS HOLDS AS LONG AS THE TROLLEY IS NOT PUSHED AGAIN.

THE DISPLACEMENT OVER THE FIRST FOUR SECONDS IS THE AREA UNDER THE CURVE, SIXTEEN METRES, BECAUSE AREA UNDER A VELOCITY-TIME GRAPH IS DISPLACEMENT, AND THIS HOLDS AS LONG AS THE VELOCITY DOES NOT CHANGE SIGN IN THAT INTERVAL.

Q4

ALL THREE TRIALS ARE RECORDED TO ONE DECIMAL PLACE, BECAUSE THE INSTRUMENT READS TO THAT PRECISION, AND THIS HOLDS AS LONG AS THE SAME INSTRUMENT IS USED THROUGHOUT.

THE TABLE SHOWS THE TIME FALLING AS THE RAMP ANGLE RISES, BECAUSE A STEEPER RAMP GIVES A LARGER ALONG-RAMP FORCE. TRIAL TWO SITS OFF THE PATTERN AND NEEDS REPEATING, BECAUSE ITS VALUE IS FAR FROM THE OTHER TWO AT THE SAME ANGLE.

Q5

C-13: B

C-14: C

C-15: A